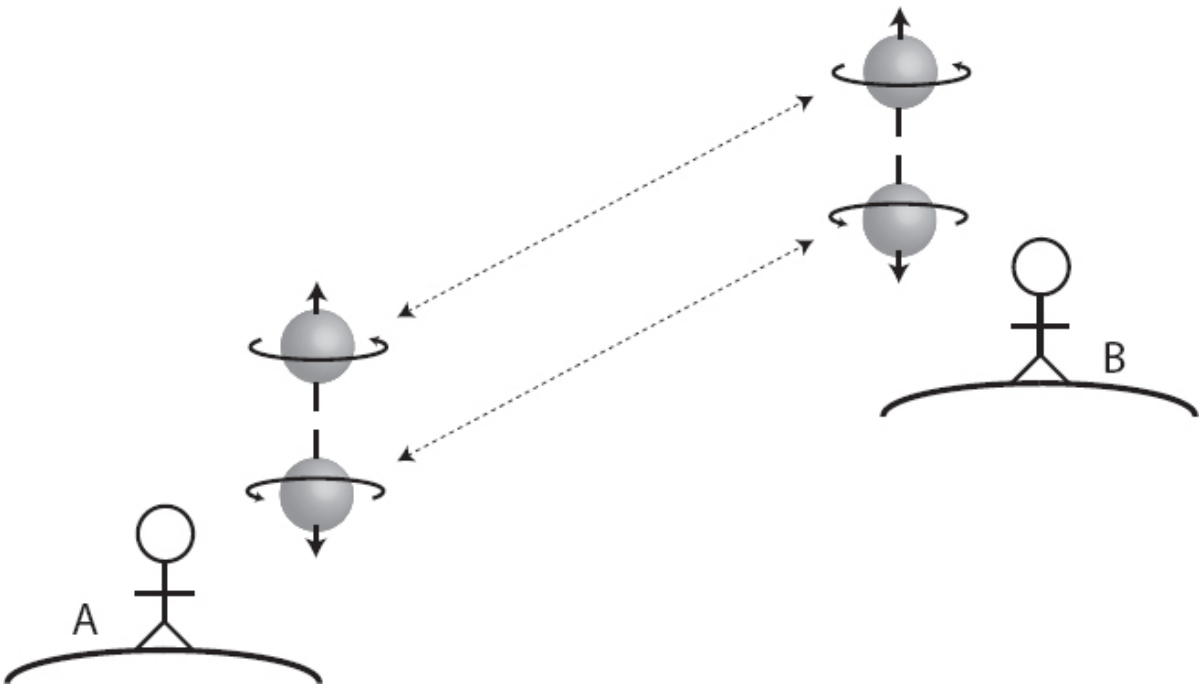


That’s weird. Thirty years earlier, Einstein had established the rules of the special theory of relativity, which says among other things that signals cannot travel faster than the speed of light. And yet here we’re saying that according to quantum mechanics, a measurement that Alice does here and now has an immediate effect on Bob’s qubit, even though it’s four light-years away. How does Bob’s qubit *know* that Alice’s has been measured, and what the outcome was? This is the “spooky action at a distance” that Einstein so memorably fretted about.

It’s not necessarily as bad as it seems. The first thing you might wonder about, upon being informed that quantum mechanics apparently sends influences faster than the speed of light, is whether or not we could take advantage of this phenomenon to communicate instantly across large distances. Can we build a quantum-entanglement phone, for which the speed of light is not a limitation at all?

No, we can’t. This is pretty clear in our simple example: if Alice measures spin-up, she instantly knows that Bob will also measure spin-up when he gets around to it. But Bob doesn’t know that. In order for him to know what the spin of his particle is, Alice has to send him her measurement result by conventional means—which are limited by the speed of light.